

Project title and goal

Title: AI-Powered Aircraft Health & Efficiency Analytics

SystemGoal:

Build a research-grade software platform that ingests open aviation data plus synthetic maintenance data to: Compute aircraft health and maintenance-risk scores Estimate fuel and CO₂ efficiency per aircraft and route Present everything in an interactive dashboard for engineering/operations users The system is a decision-support prototype, not a safety-critical or production airline tool.

2. Data sources and what they represent

Flight operations data (ADS-B / OpenSky) Track points and/or flight records for many flights (tail, origin, destination, timestamps, altitude, speed, etc.). Used to derive: Flight hours and cycles per aircraft Route distances and typical profiles Delays / irregular operations (approximate) Utilisation patterns (daily/weekly/monthly)

Weather data (METAR / basic en-route info) Weather at departure and arrival airports and simple en-route categories (clear, storms, high temperature, etc.). Used to derive: Environmental stress indicators Typical weather patterns on given routes

Engine degradation data (NASA CMAPSS) Simulated turbofan engine sensor data with Remaining Useful Life (RUL) labels. Used to: Learn generic engine health indices / RUL buckets Build features that approximate “how worn” an engine/ aircraft is Synthetic maintenance data Using rules on utilisation, RUL index, environment, and random noise to create: Scheduled maintenance events Unscheduled fault events These become labels or targets for predictive models (e.g., maintenance event in next N flights, or risk level).

No real airline-confidential data is assumed; the architecture must be easily replaceable with real data later.

3. Core features and outputs The system should produce, per aircraft (tail) and optionally per route/time window: Aircraft Health Score (0–100) Combines utilisation, degradation index (from

CMAPSS-learned models), recent anomalies and maintenance history. Maintenance Risk Level Probability of a maintenance-relevant event in the near future (e.g., next 30 days or 20 flights). Output as risk percentage + bucket (Low / Medium / High). Fuel & CO₂ Efficiency Metrics Estimated fuel burn and emissions per route and aircraft, compared against fleet average.   Efficiency index (e.g., 0–100 or % better/worse than peer fleet). Dashboard views Fleet overview: ranking of aircraft by health and risk. Aircraft detail: history of scores, utilisation, maintenance events. Route analytics: average fuel/CO₂ and risk by route, aircraft type, weather. Alerts list: aircraft/route combinations flagged as concerning.  4. Machine-learning scope At minimum, the project should implement and evaluate: Anomaly detection model Input: engineered features per aircraft over a window (utilisation, delays, weather stress, health index, etc.). Output: anomaly score or “normal / abnormal” label indicating unusual behaviour vs fleet baseline. Predictive maintenance-risk model Input: same features + recent maintenance history and degradation index. Target label: synthetic “maintenance event within next N flights/days” (Yes/No or risk level). Models to try: Logistic Regression, Random Forest / Gradient Boosting, possibly XGBoost or simple neural net. Evaluation: accuracy, precision, recall, F1, ROC-AUC on a held-out test set. Eco-efficiency estimation Regression model (or rule-based baseline) mapping flight/route features to fuel/CO₂ estimates. Optionally detect “inefficient flights” compared to model expectation.   The focus is on clean pipeline + meaningful evaluation, not on cutting-edge deep learning. 5. Architecture / components You can describe the system in these logical components: Data pipeline Scripts to download or load OpenSky flight records. Scripts to pull or mock METAR data. Loader for CMAPSS datasets. Synthetic maintenance generator module. Unified dataset builder that aggregates by

aircraft and time window. Feature engineering module Functions to compute: Flight hours, cycles, average sector length Delay metrics / irregularity metrics Weather stress scores Engine health index / RUL bucket from CMAPSS-trained model Maintenance counts and recent events Fuel and CO₂ estimates Model training & evaluation ML scripts/notebooks to: Train anomaly detection Train maintenance-risk classification/regression Train eco-efficiency model Output performance metrics, plots, and saved models. Scoring & API layer Code that loads trained models and computes scores for new data. Simple API or internal functions used by the dashboard. Dashboard (web UI) Built with Streamlit or Dash. Pages: Overview (fleet health, risk, efficiency). Aircraft detail page. Route analytics page. Alerts/what-if page where the user selects aircraft + airports + basic conditions and sees predicted scores